

Building Effective AI/BI Genie Agents

Fostering Best Practices & Lifecycle

Agenda

- Introduction & Business Value (TBA)
- Recommended Approach: A Supervised Lifecycle
- Genie AgentLifecycle
- Technical Implementation Deep Dive
- Organizational Readiness
- Scaling to Enterprise
- Future-Proofing Your Investment
- Key Takeaways & Next Steps



Recommended Approach for Engagement

A Supervised Phased Approach

Prioritize Good Topics & Build a Clear Plan

- Choose topics that are broad enough to deliver valuable insights yet narrow enough to maintain clarity.
- Develop a detailed, stakeholder-approved action plan upfront, clarifying roles, timelines, and goals.

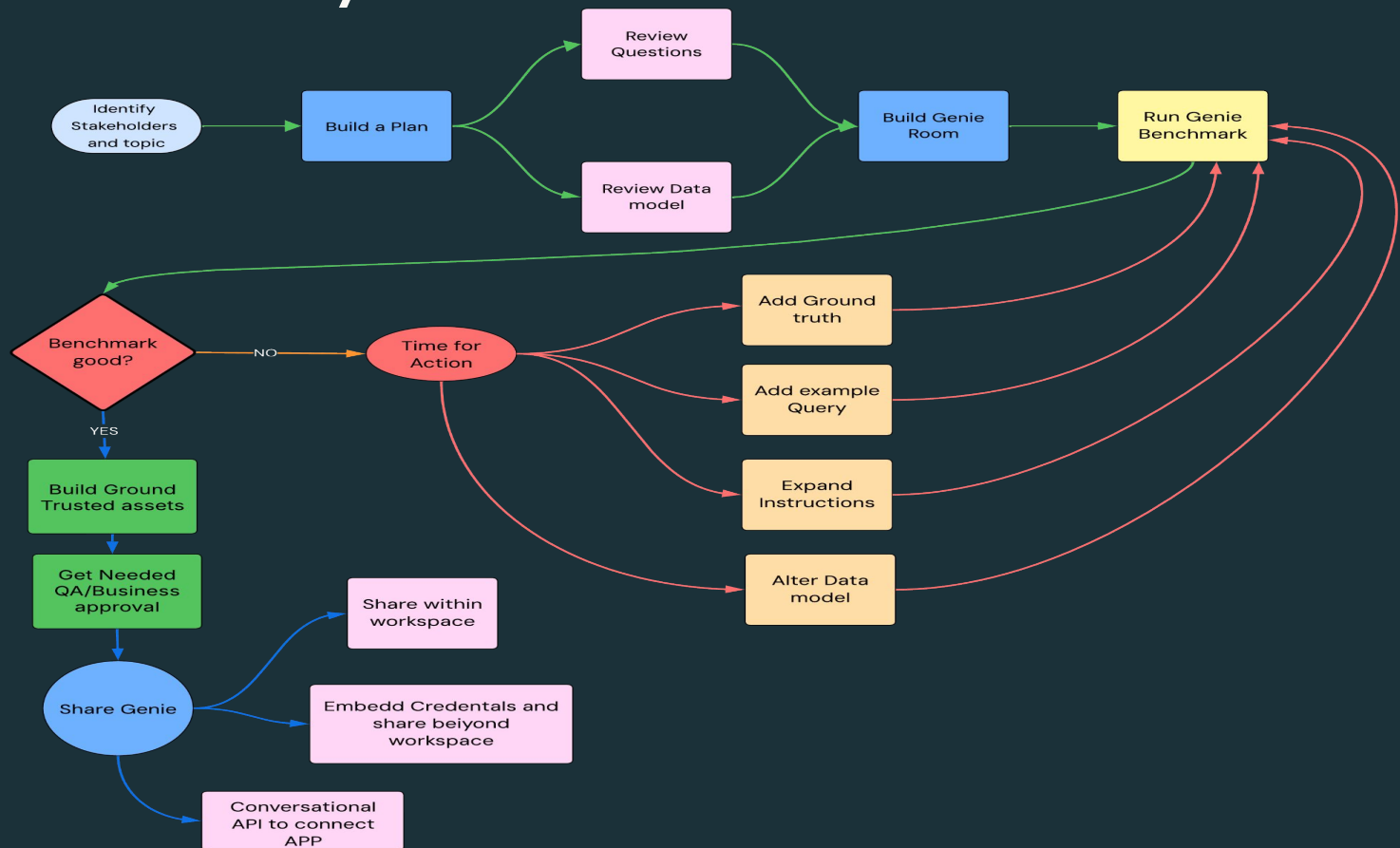
Review Questions & Validate Data Model

- Gather and prioritize specific business questions stakeholders expect Genie to address.
- Validate your data model to ensure it aligns precisely with business questions

Iterate, Maintain, and Phase Your Rollout

- Launch with a small, targeted scope first.
- Systematically iterate based on user feedback and benchmark results.
- Expand gradually in phases as accuracy and user adoption grow.

Genie LifeCycle



Stakeholder Identification and Engagement

Stakeholder Identification and Engagement



Identify Technical Stakeholders

- Own Genie Space development and maintenance.
- Understand data structures; proficient in SQL.
- Typical roles: Data Analysts, BI Developers, Tech Leads.



Identify Business Stakeholders

- End users who ask business questions.
- Provide domain knowledge and business requirements.
- Typical roles: Business Analysts, Subject Matter Experts, Sales Representatives.



Actively Engage w/t both Groups

- Conduct initial meetings to clarify scope, objectives, and expectations.
- Collect real-world questions and validate domain-specific terminology (e.g., fiscal calendar, KPIs).
- Foster collaborative feedback loops to refine the Genie Space iteratively.

Build a Plan

Define Scope & Success Metrics

- Align on target audience and business value
- Validate with both technical and business stakeholders

Create a Tactical Execution Plan

- Identify milestones, owners, and dependencies
- Include
 - Business requirements
 - Data access needs (tables, views)
 - Internal testing + signoff process
 - Timeline to get Genie into users' hands

Identify Risks & Set Timelines

- Examples:
 - Delays in stakeholder availability
 - Incomplete metadata or undefined joins
- Include dates and owners for each task

Genie Build Plan

	A	B	C	D
1	Milestone	Owner	Description	Target Date
2	Begin Plan	Databricks	Kickoff planning for Genie rollout	2025-02-02
3	Prioritize Use Case Topics	Databricks + Customer	Stack rank a list of use cases to identify the best one to begin with	2025-02-04
4	Identify Stakeholders	Customer	Identify Technical and Business stakeholders for selected use case	2025-02-07
5	Request & Review Questions	Business Stakeholder	Request business questions and review for feasibility	2025-02-09
6	Request & Review Data Model	Technical Stakeholder	Review data model with Technical Stakeholder	2025-02-12
7	Request IDP Group(s)	Technical Stakeholder	Provision access groups and permissions for Genie	2025-02-15
8	Genie Build Workshop	Databricks	Run live workshop to build and configure Genie space	2025-02-18
9	Genie Space Signoff	Technical + Business Stakeholder	Stakeholders validate Genie configuration and usability	2025-02-21
10	Genie Hack-a-thon	Technical Stakeholder	Run hackathon to test Genie with business users	2025-02-24
11	Launch to Production	Technical Stakeholder	Release Genie space to all users	2025-02-27
12	Review Usage - 1 week after launch	All Stakeholders	Review thumbs-up/down questions, impact, churn	2025-03-06
13	Review Usage - 2 weeks after launch	All Stakeholders	Second round of usage review and iteration planning	2025-03-13

Data Readiness Assessment

Data quality assessment matrix with readiness levels across different dimensions

Unity Catalog Preparation:

- **Table/Column Descriptions:** 90%+ coverage with business context
- **PK/FK Relationships:** Explicitly defined in catalog
- **Data Freshness:** Regular update schedule established

Quality Requirements:

- **Column Naming:** Consistent, descriptive conventions
- **Metadata Quality:** Business terms in descriptions
- **Sample Values:** Representative for key columns
- **Edge Cases:** NULL handling, zero values documented

Readiness Checklist:

- Curated subset of tables identified (5-10 max)
- Column descriptions added in business language
- Join paths documented for multi-table questions
- Test queries validate expected results



Review Questions & Data Model

Set a focused, realistic scope to drive successful Genie implementations.



Use Questions to Guide Space Design

- Collect stakeholder questions
- Flag common phrasing patterns (e.g., “top 10”, “last 30 days”)
- Identify frequently requested metrics/KPIs
- Clarify ambiguous terms (e.g., “churn”, “invoices”)
- Level-set unrealistic asks (e.g., “tell me something insightful”)



Curate & Validate the Data Model

- Limit to < 25 tables, ideally < 5
- Define PK/FK joins in Unity Catalog
- Use standardized naming and documented metadata
- Focus on a single grain per table (transaction-level or monthly aggregates)
- Create views to hide irrelevant columns or reduce width



Check for Gaps

- Is fiscal year logic well defined?
- Are historical date ranges covered?
- Are all needed calculations clearly defined?
- Does table metadata reflect business-friendly language?

Build the Genie Space

Make it accessible and useful

1. Configure the Space
 - Set a clear title & description
 - Select <10 relevant tables from Unity Catalog
2. Provide Context & Instructions
 - Define key business terms & logic
 - Clarify special cases (e.g., Fiscal year starts Feb)
 - Structure instructions by topic for clarity
3. Add Examples (SQL & Functions)
 - Link real user questions to example SQL
 - Parameterize frequently-used queries
 - Register SQL functions (trusted assets)
4. Interactive Testing
 - Ask realistic, typical business questions
 - Review and validate the generated SQL
 - Identify patterns where Genie struggles
 - Iteratively refine instructions & metadata

Assistant instructions
Add general instructions and example queries on how you want the space assistant to behave.

General Instructions

- win/won opportunities refer to the stagename value "5. Closed Won"
- lost opportunities refer to the stagename value "X. Closed Lost"
- An open opportunity is defined as not being "5. Closed Won" or "X. Closed Lost"
- Sales are calculated only on the opportunities where the stagename is a win/won opportunity. Sales people performance is only evaluated on their won opportunities.
- When users ask for sales reps, they want the name of the sales rep not just the ID
- Expected value of an opportunity is calculated by multiplying the probability by the amount for an opportunity.
- When an user asks about past purchases, check the opportunityhistory for that customer and the number of won opportunity updates there are

Join relationships:

- opportunity.id is the primary key, opportunityhistory.opportunityid is the foreign key
- accounts.id is the primary key, opportunity.accountid is the foreign key. A customer's region info is in the accounts table.

Discard Save

Example SQL Queries
Add example queries that the assistant can learn from.

+ Add example query

what were the biggest closed deals in 2023 and who were the sales reps for them?

```
SELECT o.name AS deal_name, u.name AS sales_rep_name, o.amount, o.cl...
```

which sales reps converted open opportunities to closed opportunities at the highest percentage?

```
SELECT u.name AS sales_rep_name, COUNT(o.id) AS total_opportunities,...
```

Data
The tables power the space. Select the tables that you want to give as context to the space in the [edit form](#).

>  accounts
This table contains information about accoun...

<  opportunity
This table contains the most updated informa...
[Open in Catalog Explorer](#)

 id
the unique ID identifier for an opportunity

 ownerid
the sales person who owns this opportunity

 stagename
the current stage of the opportunity. Currentl...

 accountid
No comment available

 type
No comment available

the current stage of the opportunity. Currently used values are: "1. Discovery", "2. Demo", "3. Validation", "4. Procure", "5. Closed Won", "X. Closed Lost".

Run Genie Benchmark

Step 1: Create Benchmark Questions

- Collect frequently asked questions from business users
- Include variations in phrasing to test Genie's flexibility
- (Recommended) Provide corresponding "ground truth" SQL queries

Step 2: Execute & Evaluate

- Run benchmarks regularly to measure Genie's accuracy
- Review benchmark labels:
 - **Good:** Accurate responses matching ground truth
 - **Manual Review Needed:** Potential mismatches or unclear results

Step 3: Take Action

- Mark each response clearly as **Good** or **Bad** after review
 - Identify common issues or recurring errors
- Update instructions, examples, or simplify data structure based on results

Add benchmark

Question

e.g. "What was our ARR over the last 4 years?"

SQL Answer (optional)

▶ Run

Enter a ground truth SQL statement that correctly answers your benchmark question. This statement will be executed, and the query results will be compared to Genie's results to determine accuracy. Questions without a ground truth SQL statement will be marked for manual review.

CancelAdd benchmark

Evaluations

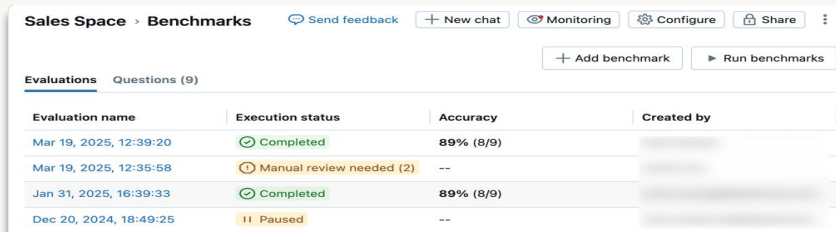
2024-10-06 06:56:25

Assessment completeManual review needed (23)

What is the monthly DBU usage and...	What is the monthly DBU usage and total Databricks cost for tag project = dbdemos and tag demo = auto_loader from January until now
What are the top 10 most expensive p...	Assessment: Good
Please identify the top 10 most...	Question
Please identify sources of DBU usage...	What is the monthly DBU usage and total Databricks cost for tag project = dbdemos and tag demo = auto_loader from January until now
What is the cost of this vector search...	Response
What are the top 15 most expensive...	Model output SQL
What is the total usage and databricks...	Ground truth
What are the top 15 most expensive...	No benchmark SQL added
For the cluster that starts with...	
What are the top 15 most expensive s...	
For project tag dbdemos, what are th...	
For project tag dbdemos, what are th...	

Assess Benchmark Results

Evaluate performance and decide your next course of action



The screenshot shows the 'Sales Space > Benchmarks' interface. It includes a top navigation bar with 'Send feedback', 'New chat', 'Monitoring', 'Configure', and 'Share'. Below this is a sub-header with 'Evaluations' and 'Questions (9)', along with '+ Add benchmark' and 'Run benchmarks' buttons. The main content is a table with four columns: 'Evaluation name', 'Execution status', 'Accuracy', and 'Created by'. The table contains four rows of data.

Evaluation name	Execution status	Accuracy	Created by
Mar 19, 2025, 12:39:20	Completed	89% (8/9)	
Mar 19, 2025, 12:35:58	Manual review needed (2)	--	
Jan 31, 2025, 16:39:33	Completed	89% (8/9)	
Dec 20, 2024, 18:49:25	Paused	--	

If Results Are Accurate

- Proceed to create trusted assets (views, precomputed metrics).
- Validate with business stakeholders.
- Move to broader rollout and sharing.

If Results Are Inaccurate

Identify root causes:

- Missing or unclear metadata?
- Complex business logic not yet explained?
- Incorrect joins or filters?

Use this insight to plan refinements:

- Update instructions or metadata.
- Add ground truth or example queries.
- Revisit data model if needed.

Taking Action to Make Benchmarks better

#	Approach	How to implement	Consideration for Implementation
1	Add a Ground Truth SQL for question which needs review	- For each question needs review, identify the SQL, run in SQL editor, validate the results. - Put the right SQL in Ground Truth after testing the results	- Easy to implement - Use this as first step to improve benchmark
2	Add Example Query for mostly asked questions	- Find commonly asked questions by working with business stakeholders. - Add example query and questions for commonly asked questions	- Easy to implement - Use this if there are many benchmarks needing updates to ground truth
3	Expand Instructions to guide genie on how to response	Add context and response behavior in the query e.g. adding acronyms or KPI calculations in instructions	- Easy but need user with business and SQL understanding - Use this if example queries are not sufficient or need genie to redirect to SQL function or use company acronym etc
4	Revalidate Data Model	- Find all benchmarks which is affected by missing date - Review the data and either enhance model to support question or communicate users about the scope	- Little complex and need users who understand backend data - Use this if most of the queries are not able to return data or results with previous approaches
5	Add SQL Function(table/UDF) to expand reusability	- This steps enhance reusability to use same calculation or query logic across multiple rooms - Add instruction to call SQL Functions for specific criteria	- Little complex and need user with good understanding of data and SQL - Use this for reusability so multiple rooms can response rightly for similar questions



Build Trusted Assets

- **Target recurring, high-value questions**
Ideal for known metrics, KPIs, and commonly asked business queries.
- **Use parameterized SQL queries or UDFs**
Promote reusability and clarity; define in Unity Catalog.
- **Add rich comments in functions**
Describe intent, input format, and examples to help Genie trigger the asset correctly.
- **Keep logic clean and scoped**
Avoid overly complex joins or business rules Genie can't infer.
- **Organize in a dedicated schema**
Simplifies access and permissioning (CAN USE, EXECUTE).
- **Maintain as governed assets**
Treat as contract-driven logic: review regularly with domain owners.

What are my open opportunities in EMEA



Genie



Trusted

Found a function that might match your question: `open_opps_in_region` ⓘ

▼ Show less

Addresses questions about the pipeline in the specified regions by returning a list of all the open opportunities. If no region is specified, returns all open opportunities. Example questions: "What is the pipeline for APAC and EMEA?", "Open opportunities in APAC"

• regions: EMEA

SQL

```
regions ARRAY < STRING > COMMENT 'List of regions. Values: ["AF","AN","AS", "EU", "NA", "OC", "SA", NULL]'
```

SQL

```
min_date STRING COMMENT 'minimum date (included) for a transaction, in 'yyyy-mm-dd' format'
```

SQL

```
CREATE OR REPLACE FUNCTION my_catalog.my_schema.calculate_bmi(weight_kg DOUBLE, height_m DOUBLE)
RETURNS DOUBLE
COMMENT "Calculates Body Mass Index (BMI) from weight and height."
LANGUAGE PYTHON
AS $$
"""
Parameters:
calculate_bmi (version 1.2):
- weight_kg (float): Weight of the individual in kilograms.
- height_m (float): Height of the individual in meters.

Returns:
- float: The calculated BMI.

Example Usage:
SELECT calculate_bmi(weight, height) AS bmi FROM person_data;

Change Log:
- 1.0: Initial version.
- 1.1: Improved error handling for zero or negative height values.
- 1.2: Optimized calculation for performance.

Note: BMI is calculated as weight in kilograms divided by the square of height in meters.
"""
if height_m <= 0:
    return None # Avoid division by zero and ensure height is positive
return weight_kg / (height_m ** 2)
$$
```

Change Management & User Adoption

Stakeholder Identification and Engagement

Driving Successful Adoption:

- Executive Sponsorship: Secure visible leadership support
- Success Stories: Showcase early wins and time savings
- Training Program: Role-based onboarding sessions
- Champions Network: Identify and empower power users

Overcoming Resistance:

- Address "trust gap" with benchmark demonstrations
- Provide side-by-side comparisons with manual SQL
- Create feedback channels for improvement suggestions
- Celebrate and reward active users

Measuring Adoption:

- Daily/weekly active users
- Questions asked per user/team
- Feedback ratings (thumbs up/down)
- Reduction in ad-hoc query requests to data team

Change Management & User Adoption

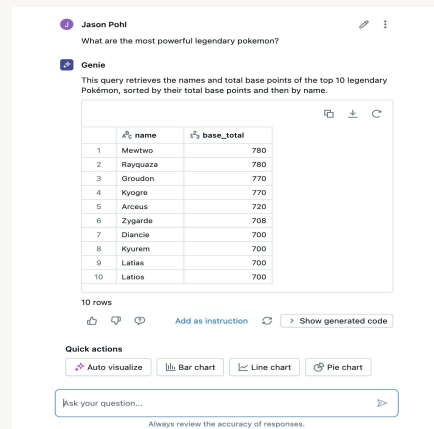
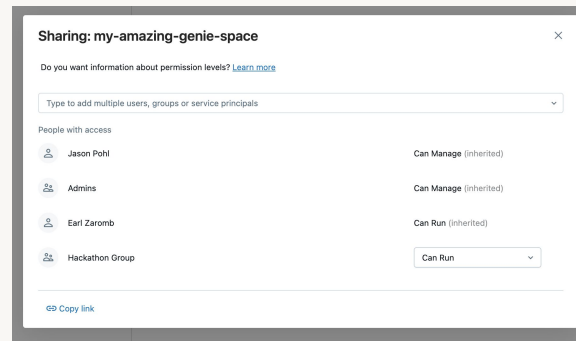
Make it accessible and useful



Sharing & Scaling Genie Spaces

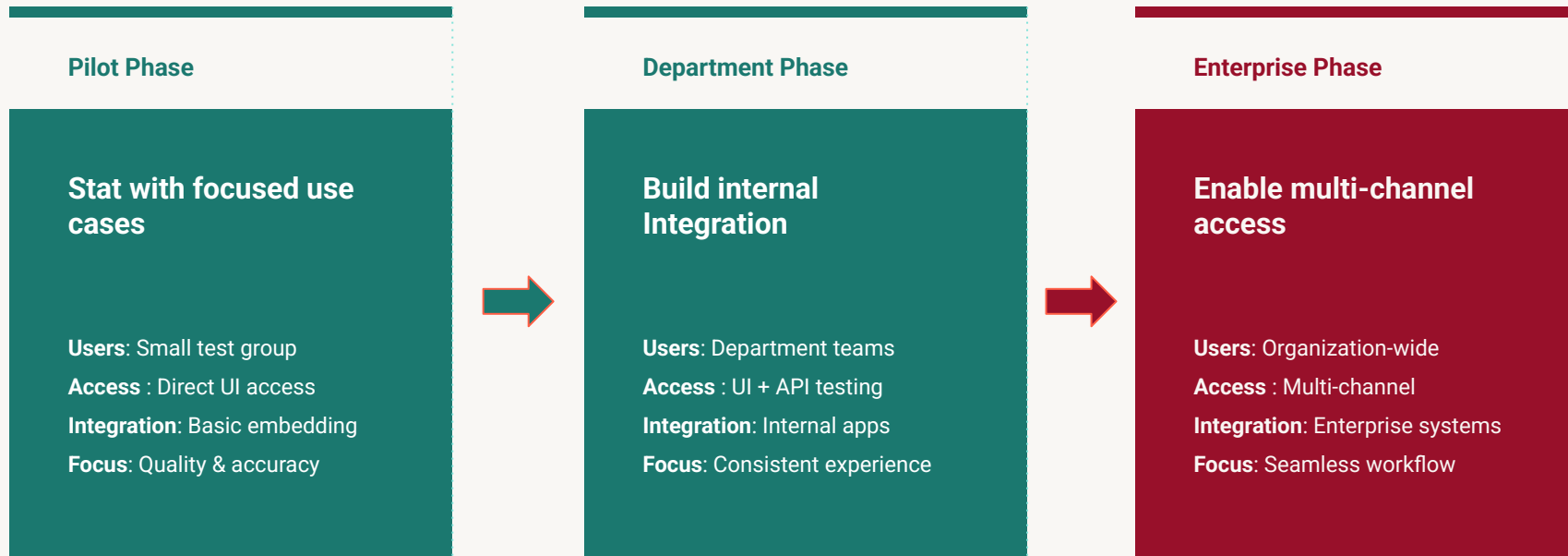
Make it accessible and useful

- **Start with a small user group**
 - Test with a focused audience before scaling more broadly.
- **Set clear permissions**
 - Use Unity Catalog to control access (**CAN USE**, **CAN EDIT**, **EXECUTE**).
- **Provide lightweight onboarding**
 - Share quick-start guides or sample questions to help users engage effectively.
- **Monitor usage & feedback**
 - Use the History tab and user ratings to spot trends and improve coverage.
- **Clone for new domains or teams**
 - Reuse proven spaces as templates for similar use cases with minimal rework.



Scaling to Enterprise

Phases with increasing scope and integration complexity



Integration Options : Conversation API, Microsoft Teams, Custom Portals, Multi-Agent Systems

Ongoing Monitoring & Evolution

Continuously improve based on real-world usage and feedback

- **Review usage regularly**
 - Use the History tab to track common questions and coverage gaps.
- **Refine instructions & assets iteratively**
 - Adjust metadata, expand instructions, or add trusted assets based on actual queries.
- **Re-run benchmark tests periodically**
 - Validate that Genie still answers key questions accurately as data or logic changes.
- **Maintain stakeholder feedback loops**
 - Encourage users to upvote/downvote and share missed expectations.
- **Plan for ongoing governance**
 - Set review cadences for trusted assets, access controls, and data quality checks.

The screenshot shows the 'Monitoring' tab of a Genie Space. It displays a list of questions asked by users, their status, ratings, and creation times. The interface includes filters for 'Last 30 days', 'Rating', 'User', and 'Status'.

Status	Question	Rating	User	Created
Completed	what are all my opportunities that started in december fy 2023?		Hanlin Sun	2024-05-28 15:4...
Completed	what about just in December FY 2023?		Hanlin Sun	2024-05-28 15:3...
Completed	which sales reps converted open opportunities to closed opportunities at the highest percentage?		Hanlin Sun	2024-05-28 15:3...
Completed	which quarter in 2023 saw the most opportunities go from open to won?		Hanlin Sun	2024-05-28 15:3...
Completed	what were the biggest closed deals in 2023 and who were the sales reps for them?		Hanlin Sun	2024-05-28 15:2...
Completed	what were the biggest closed deals in 2023 and who were the sales reps for them?		Hanlin Sun	2024-05-28 15:1...
Completed	What is the expected value of all open opportunities in Dec 2023?		Hanlin Sun	2024-05-28 15:1...
Completed	what is the most recent sales opportunity?		Hanlin Sun	2024-05-28 15:0...
Completed	what is the average?		Hanlin Sun	2024-05-28 14:5...
Completed	what is the smallest?		Hanlin Sun	2024-05-28 14:5...
Completed	what is the biggest sales opportunity i've ever had?		Hanlin Sun	2024-05-28 14:5...
Completed	What is the expected value of all open opportunities in Dec 2023?		chao.cai@databricks.com	2024-05-28 14:1...

The screenshot shows the 'History' tab of a Genie Space. It displays a list of activities, including questions, ratings, requests, comments, users, creation times, and last update times. The interface includes filters for 'Last 30 days', 'Rating', 'Request', 'User', 'Status', and 'Reset filters'.

Question	Rating	Request	Comment	User	Created	Last updated
Yes		Requested	We should create ...	Jason Pohl	2024-12-17 15:17...	2024-12-17 15:18:11

Key Takeaways

- Start small with curated data and a clear use case
- Teach Genie with precise instructions and trusted assets
- Use benchmarks and real questions to validate accuracy
- Iterate based on feedback and evolve continuously
- Scale confidently by cloning proven spaces and monitoring usage
- Common Pitfalls to Avoid
 - Adding too many tables initially
 - Skipping benchmark testing
 - Neglecting change management
 - Underinvesting in instructions and examples



Q&A



Organizational Readiness

Matrix showing readiness dimensions across different organizational roles



Skills Assessment

- Data Team: SQL expertise, Unity Catalog knowledge
- Business Users: Domain knowledge, question formulation
- IT/Security: Access management, governance



Team Structure

- Genie Space Owner: Overall accountability
- Content Developers: Instructions, examples, functions
- Subject Matter Experts: Domain validation
- Executive Sponsor: Resource allocation, visibility



Governance Framework

- Clear ownership of spaces and content
- Review process for instructions and trusted assets
- Usage monitoring and access controls
- Compliance documentation and audit trails

Appendix

